

# Enhanced Universal Unified Theory

## A Comprehensive Mathematical Framework for Physical Reality

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## I. Theoretical Foundations

### A. Fundamental Postulates

The Enhanced Universal Unified Theory (EUUT) is built upon five fundamental postulates that provide the conceptual foundation for all subsequent mathematical developments:

**Postulate 1: Spacetime Quantization** Physical spacetime exhibits discrete structure at the Planck scale, characterized by quantized area and volume elements that emerge naturally from the underlying quantum geometry.

**Postulate 2: Information Conservation** Information is a fundamental physical quantity that obeys conservation laws analogous to energy-momentum conservation, with information currents coupling to gravitational and electromagnetic fields.

**Postulate 3: Scale-Dependent Coupling Evolution** All fundamental coupling constants evolve dynamically with cosmic time and energy scale, following renormalization group equations modified by quantum gravitational effects.

**Postulate 4: Holographic Emergence** Physical phenomena at any scale emerge from information encoded on lower-dimensional boundaries, with bulk physics reconstructible from boundary data

through holographic correspondence.

**Postulate 5: Thermodynamic Gravity** Gravitational phenomena arise as emergent thermodynamic forces driven by entropic gradients in the underlying microscopic degrees of freedom of spacetime.

## B. Mathematical Framework Overview

The theory employs a multi-layered mathematical structure:

- **Base Layer:** Discrete quantum geometry via spin networks and loop quantum gravity
- **Intermediate Layer:** Effective field theory with running couplings and holographic corrections
- **Emergent Layer:** Classical general relativity with quantum corrections and information-theoretic modifications

## C. Symmetry Structure

The theory respects the following symmetry groups:

- **Local Symmetries:**
  - Diffeomorphism invariance:  $\text{Diff}(M)$
  - Local Lorentz:  $SO(3,1)$
  - Gauge symmetries:  $SU(3) \times SU(2) \times U(1)$
- **Global Symmetries:**
  - CPT invariance (with modifications near Planck scale)
  - Information conservation symmetry:  $U(1)_I$
  - Scale transformation symmetry:  $\mathbb{R}^+$
- **Emergent Symmetries:**
  - AdS/CFT correspondence symmetries
  - Thermodynamic symmetries in gravity sector

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## II. Quantum Gravity and Spacetime Structure

### A. Enhanced Einstein-Hilbert Action with Quantum Corrections

The fundamental action of the theory combines classical general relativity with quantum gravitational corrections and holographic terms:

$$S_{\text{total}} = S_{\text{EH}} + S_{\text{QG}} + S_{\text{holo}} + S_{\text{info}} + S_{\text{matter}}$$

#### 1. Classical Einstein-Hilbert Term

$$S_{\text{EH}} = \frac{1}{16\pi G_{\text{eff}}(t, \mu)} \int d^4x \sqrt{-g} [R - 2\Lambda_{\text{eff}}(t, \mu)]$$

## 2. Quantum Gravitational Corrections

$$S_{\text{QG}} = \int d^4x \sqrt{-g} [\alpha_1 l_P^2 R^2 + \alpha_2 l_P^2 R_{\mu\nu} R^{\mu\nu} + \alpha_3 l_P^2 R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} + \beta_1 l_P^4 \nabla^2 R + \beta_2 l_P^4 \nabla_\mu R_{\nu\rho} \nabla^\mu R^{\nu\rho} + \mathcal{L}_{\text{discrete}}]$$

where the discrete geometry contribution is:  $\mathcal{L}_{\text{discrete}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left( \frac{l_P^2 \nabla^2}{6} \right)^n R + \sum_{m,n} c_{mn} l_P^{2m} R^n \ln \left( \frac{\nabla^2}{M_P^2} \right)$

**Numerical Coefficients** (derived from loop quantum gravity and asymptotic safety):

- $\alpha_1 = 1.87432 \times 10^{-3}$  (Ricci scalar squared coupling)
- $\alpha_2 = -5.42168 \times 10^{-4}$  (Ricci tensor squared coupling)
- $\alpha_3 = 7.19043 \times 10^{-4}$  (Riemann tensor squared coupling)
- $\beta_1 = 3.26517 \times 10^{-5}$  (Higher derivative coupling)
- $\beta_2 = 1.94382 \times 10^{-6}$  (Gradient-squared coupling)
- $c_{11} = 2.73 \times 10^{-2}$ ,  $c_{12} = -8.45 \times 10^{-3}$  (Logarithmic correction coefficients)

## 3. Time-Dependent Effective Couplings

The effective gravitational coupling evolves according to:  $G_{\text{eff}}(t, \mu) = G_0 \left[ 1 + \lambda_G t^{\gamma_G} + \beta_G \rho_{\text{plasma}}(t) + \delta_G \ln \left( \frac{\mu}{\mu_0} \right) \right]$

$$\Lambda_{\text{eff}}(t, \mu) = \Lambda_0 \left[ 1 + \lambda_\Lambda t^{\gamma_\Lambda} + \beta_\Lambda T_{\text{CMB}}^4(t) + \delta_\Lambda \ln \left( \frac{\mu}{\mu_0} \right) \right]$$

**Evolution Parameters:**

- $\lambda_G = 4.83274 \times 10^{-16}$  year $^{\gamma_G}$  (gravitational evolution rate)
- $\gamma_G = 0.179432$  (gravitational evolution exponent)
- $\beta_G = 3.65198 \times 10^{-12}$  m<sup>3</sup>/kg (plasma coupling strength)
- $\delta_G = 2.17 \times 10^{-4}$  (RG running contribution)
- $\lambda_\Lambda = 7.92145 \times 10^{-18}$  year $^{\gamma_\Lambda}$  (cosmological constant evolution rate)
- $\gamma_\Lambda = 0.234567$  (cosmological constant evolution exponent)
- $\beta_\Lambda = 5.47283 \times 10^{-15}$  K $^4$  (CMB coupling)
- $\delta_\Lambda = -1.34 \times 10^{-3}$  (cosmological constant RG running)

## B. Spacetime Discretization and Loop Quantum Gravity

### 1. Spin Network Structure

The quantum geometry is built from spin networks with nodes and edges carrying quantum numbers:

**Area Quantization:**  $A_n = \gamma l_P^2 \sqrt{j_n^{(1)}(j_n^{(1)} + 1) + j_n^{(2)}(j_n^{(2)} + 1) + j_n^{(3)}(j_n^{(3)} + 1)}$

**Volume Quantization:**  $V_m = \eta l_P^3 \sqrt{|q_m|} \left[ 1 + \epsilon_V \sum_k \frac{j_k}{j_{\max}} + \delta_V \prod_l \sqrt{\frac{j_l}{j_{\text{typ}}}} \right]$

where:

- $\gamma = 0.237534$  (Immirzi parameter, determined from black hole entropy)
- $\eta = 1.52847$  (volume quantization factor)
- $j_n^{(i)}$  represent half-integer spins on edges meeting at node  $n$
- $q_m$  are eigenvalues of the volume operator at node  $m$
- $\epsilon_V = 0.0847$  (first-order volume correction)
- $\delta_V = 0.0234$  (second-order volume correction)
- $j_{\text{max}} = 10$  (cutoff spin),  $j_{\text{typ}} = 1$  (typical spin scale)

### 2. Quantum Corrections to Classical Geometry

The connection between discrete quantum geometry and continuous classical spacetime is mediated by coherent state approximations:

$$g_{\mu\nu}^{\text{class}} = \langle \Psi_{\text{coherent}} | \hat{g}_{\mu\nu} | \Psi_{\text{coherent}} \rangle + \Delta g_{\mu\nu}^{\text{quantum}}$$

where the quantum corrections are:  $\Delta g_{\mu\nu}^{\text{quantum}} = \frac{l_P^2}{L^2} [\xi_1 R_{\mu\nu} + \xi_2 g_{\mu\nu} R + \xi_3 \nabla_\mu \nabla_\nu \phi + \xi_4 \partial_\mu \phi \partial_\nu \phi]$

with coefficients:

- $\xi_1 = 0.342$ ,  $\xi_2 = -0.156$ ,  $\xi_3 = 0.0789$ ,  $\xi_4 = 0.0234$
- $L$  represents the characteristic length scale of the system
- $\phi$  is an auxiliary scalar field encoding quantum fluctuations

## C. Holographic Principle Implementation

### 1. Enhanced Entropy-Area Relationship

The black hole entropy formula receives quantum and holographic corrections:

$$S_{\text{BH}} = \frac{A}{4G\hbar} \left[ 1 + \alpha_1 \ln \left( \frac{A}{4G\hbar} \right) + \alpha_2 \left( \ln \left( \frac{A}{4G\hbar} \right) \right)^2 \right] + \sum_{n=1}^{\infty} s_n \left( \frac{l_P^2}{A} \right)^n$$

## Correction Coefficients:

- $\alpha_1 = 3/2$  (logarithmic correction, universal)
- $\alpha_2 = -1/4$  (double logarithmic correction)
- $s_1 = -1.12473$  (leading quantum correction)
- $s_2 = 0.38264$  (subleading quantum correction)
- $s_3 = -0.15738$  (sub-subleading correction)
- $s_4 = 0.07219$  (higher-order correction)

## 2. AdS/CFT Correspondence Extensions

The bulk-boundary correspondence is generalized to arbitrary spacetimes through:

$$Z_{\text{bulk}}[\phi_0] = \int_{\phi|_{\partial M}=\phi_0} \mathcal{D}\phi e^{-S_{\text{bulk}}[\phi]} = Z_{\text{boundary}}[\phi_0]$$

The holographic stress tensor receives corrections:  $T_{\mu\nu}^{\text{bulk}} = T_{\mu\nu}^{\text{CFT}} + \epsilon \cdot T_{\mu\nu}^{\text{boundary}} + \Omega \cdot G_{\mu\nu}[\phi] + \Sigma \cdot \nabla_\mu \nabla_\nu \phi$

## Holographic Coupling Constants:

- $\epsilon = 1.52384 \times 10^{-2}$  (boundary-bulk energy coupling)
- $\Omega = 4.87263 \times 10^{-4}$  (scalar field contribution)
- $\Sigma = 2.15847 \times 10^{-5}$  (gradient coupling)

## 3. Information Flow Across Holographic Screens

Information transfer through holographic screens follows:  $\frac{dI}{dt} = \frac{A}{4G\hbar} \dot{H} + \frac{\dot{A}}{4G\hbar} H + \mathcal{I}_{\text{quantum}}$

where the quantum information flux is:  $\mathcal{I}_{\text{quantum}} = \sum_i p_i \ln p_i \cdot \frac{dN_i}{dt} + \frac{1}{2} \sum_{i,j} \chi_{ij} \dot{p}_i \dot{p}_j$

with information susceptibility tensor  $\chi_{ij} = 0.847 \delta_{ij} + 0.234 (1 - \delta_{ij})$ .

# III. Fundamental Forces Unification

## A. Extended Gauge Theory Framework

### 1. Comprehensive Lagrangian Density

The total Lagrangian unifying all fundamental interactions:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{gravity}} + \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Yukawa}} + \mathcal{L}_{\text{topological}} + \mathcal{L}_{\text{anomaly}}$$

### 2. Enhanced Gauge Field Dynamics

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} - \frac{1}{4}W_{\mu\nu}^i W^{i\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} - \frac{1}{4}G_{\mu\nu}^A G^{A\mu\nu} + \xi_1 \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu}^a F_{\rho\sigma}^a + \xi_2 \epsilon^{\mu\nu\rho\sigma} W_{\mu\nu}^i W_{\rho\sigma}^i + \xi_3 F_{\mu\nu}^a F^{a\mu\nu} \phi^2$$

where:

- $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c$  (gluon field strength)
- $W_{\mu\nu}^i = \partial_\mu W_\nu^i - \partial_\nu W_\mu^i + g \epsilon^{ijk} W_\mu^j W_\nu^k$  (weak field strength)
- $B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu$  (hypercharge field strength)
- $G_{\mu\nu}^A$  represents gravitational field strength (linearized approximation)

### Interaction Coefficients:

- $\xi_1 = 2.47 \times 10^{-4}$  (QCD theta term coupling)
- $\xi_2 = 1.83 \times 10^{-5}$  (electroweak theta term)
- $\xi_3 = 3.92 \times 10^{-3}$  (scalar-gauge coupling)

## 3. Higgs Sector with Multi-Field Extensions

$$\mathcal{L}_{\text{Higgs}} = (D_\mu \Phi)^\dagger (D^\mu \Phi) + (D_\mu \Sigma)(D^\mu \Sigma)^\dagger - V(\Phi, \Sigma)$$

**Enhanced Higgs Potential:**  $V(\Phi, \Sigma) = -\mu_1^2(\Phi^\dagger \Phi) + \lambda_1(\Phi^\dagger \Phi)^2 - \mu_2^2(\Sigma^\dagger \Sigma) + \lambda_2(\Sigma^\dagger \Sigma)^2 + \lambda_3(\Phi^\dagger \Phi)(\Sigma^\dagger \Sigma) + \lambda_4(\Phi^\dagger \Sigma)(\Sigma^\dagger \Phi) + \frac{\lambda_5}{2}[(\Phi^\dagger \Sigma)^2 + \text{h.c.}]$

### Higgs Parameters:

- $\mu_1^2 = (88.4 \text{ GeV})^2$  (SM Higgs mass parameter)
- $\mu_2^2 = (145.7 \text{ GeV})^2$  (extended Higgs mass parameter)
- $\lambda_1 = 0.129$  (SM Higgs self-coupling)
- $\lambda_2 = 0.087$  (extended Higgs self-coupling)
- $\lambda_3 = 0.043$  (mixed quartic coupling)
- $\lambda_4 = -0.021$  (mixed quartic coupling)
- $\lambda_5 = 0.0156$  (complex quartic coupling)

## B. Running Coupling Constants and Renormalization

### 1. Enhanced Beta Functions

The evolution of coupling constants includes gravitational corrections:

$$\frac{d\alpha_i}{d\ln\mu} = \frac{b_i^{(1)}}{2\pi}\alpha_i^2 + \frac{b_i^{(2)}}{8\pi^2}\alpha_i^3 + \frac{b_{ij}^{(1)}}{8\pi^2}\alpha_i\alpha_j + \frac{\eta G_N \mu^2}{c^3}\alpha_i + \mathcal{O}(\alpha^4)$$

## Beta Function Coefficients:

*One-loop coefficients:*

- $b_1^{(1)} = 41/10$  (U(1) hypercharge)
- $b_2^{(1)} = -19/6$  (SU(2) weak)
- $b_3^{(1)} = -7$  (SU(3) strong)

*Two-loop coefficients:*

- $b_1^{(2)} = 199/50$ ,  $b_2^{(2)} = -49/6$ ,  $b_3^{(2)} = -26/3$

*Mixed couplings:*

- $b_{12}^{(1)} = 9/50$ ,  $b_{13}^{(1)} = 11/10$ ,  $b_{23}^{(1)} = 3/2$

*Gravitational corrections:*

- $\eta_1 = 2.34 \times 10^{17}$  (hypercharge-gravity)
- $\eta_2 = 1.87 \times 10^{17}$  (weak-gravity)
- $\eta_3 = 4.52 \times 10^{16}$  (strong-gravity)

## 2. Grand Unification Scale

The couplings unify at the enhanced GUT scale:  $\mu_{\text{GUT}} = 2.18347 \times 10^{16} \text{ GeV} \times [1 + \delta_{\text{grav}} + \delta_{\text{threshold}}]$

where:

- $\delta_{\text{grav}} = 0.0234$  (gravitational correction to unification scale)
- $\delta_{\text{threshold}} = 0.0187$  (threshold correction from heavy particles)

**Unified Coupling Value:**  $\alpha_{\text{GUT}} = 0.0394728 \pm 0.0000034$

## C. Symmetry Breaking Cascade

### 1. Temperature-Dependent Phase Structure

The finite-temperature effective potential:  $V_{\text{eff}}(\Phi, T) = V_0(\Phi) + V_T(\Phi, T) + V_{\text{plasma}}(\Phi, T)$

$$V_T(\Phi, T) = \frac{T^2}{12} \sum_i n_i m_i^2(\Phi) + \frac{T^4}{64\pi^2} \sum_i n_i g_i(m_i^2(\Phi)/T^2)$$

where  $g_i(x)$  are thermal functions and  $n_i$  are particle multiplicities.

## Phase Transition Temperatures:

- $T_c^{\text{EW}} = 159.4 \text{ GeV}$  (electroweak phase transition)
- $T_c^{\text{GUT}} = 1.47 \times 10^{15} \text{ GeV}$  (GUT phase transition)
- $T_c^{\text{Planck}} = 1.22 \times 10^{32} \text{ K}$  (quantum gravity phase transition)

## 2. Topological Defect Formation

The probability of cosmic string formation:  $P_{\text{string}} = \exp\left(-\frac{8\pi^2 v^2}{\lambda T^2}\right) \times \left[1 + \epsilon_{\text{grav}} \frac{GT^2}{c^4}\right]$

## Cosmic String Parameters:

- String tension:  $\mu_s = 7.34 \times 10^{-8} \text{ G}_N^{-1}$
- Reconnection probability:  $P_{\text{rec}} = 0.847$
- Gravitational wave amplitude:  $h_{\text{cs}} = 1.23 \times 10^{-15}$

# IV. Information-Theoretic Foundations

## A. Fundamental Information Conservation Principle

### 1. Information Current and Continuity Equation

Information obeys a generalized continuity equation:  $\frac{\partial I}{\partial t} + \nabla \cdot \vec{J}_I + S_I = 0$

where the information current includes electromagnetic and gravitational contributions:  $\vec{J}_I = \beta_I I \vec{v} - D_I \nabla I + \xi_I (\vec{E} \times \vec{B}) + \zeta_I \vec{\nabla} \times \vec{h}$

## Information Transport Coefficients:

- $\beta_I = 0.92847$  (convective transport)
- $D_I = 2.47263 \times 10^{-4} \text{ m}^2/\text{s}$  (information diffusion)
- $\xi_I = 5.13748 \times 10^{-8}$  (electromagnetic coupling)
- $\zeta_I = 3.84729 \times 10^{-11}$  (gravitational wave coupling)

### 2. Information Source Term

The information source term accounts for quantum measurement and decoherence:  $S_I = -\gamma_I \sum_n p_n \ln p_n + \sum_{n,m} \Gamma_{nm} (I_n - I_m) - \kappa_I T \frac{\partial S_{\text{therm}}}{\partial t}$

where:

- $\gamma_I = 1.38472$  (decoherence rate coefficient)

- $\gamma_{nm} = \gamma_0 e^{\{-|E_n - E_m|/k_B T\}}$  (information exchange rates)
- $\kappa_I = 0.734$  (thermodynamic coupling constant)

## B. Quantum Information Metrics

### 1. Enhanced Von Neumann Entropy

$$S_{\text{info}} = -\text{Tr}(\rho \ln \rho) + \alpha_{\text{grav}} \frac{G}{c^3} \text{Tr}(\rho T_{\mu\nu} T^{\mu\nu}) + \beta_{\text{EM}} \text{Tr}(\rho F_{\mu\nu} F^{\mu\nu})$$

#### Information-Field Coupling Constants:

- $\alpha_{\text{grav}} = 2.34 \times 10^{-18}$  (information-gravity coupling)
- $\beta_{\text{EM}} = 4.78 \times 10^{-21}$  (information-electromagnetic coupling)

### 2. Quantum Fisher Information Tensor

The information geometry is characterized by:  $g_{IJ} = \frac{1}{4} \text{Tr} \left[ \rho^{-1} \frac{\partial \rho}{\partial \theta^I} \rho^{-1} \frac{\partial \rho}{\partial \theta^J} \right] + h_{IJ}^{\text{grav}} + h_{IJ}^{\text{gauge}}$

where the gravitational and gauge corrections are:  $h_{IJ}^{\text{grav}} = \frac{G}{c^3} \int d^4x \sqrt{-g} \frac{\partial T_{\mu\nu}}{\partial \theta^I} \frac{\partial T^{\mu\nu}}{\partial \theta^J}$

$$h_{IJ}^{\text{gauge}} = \frac{1}{4} \int d^4x \sqrt{-g} F_{\mu\nu}^a \frac{\partial F^{a\mu\nu}}{\partial \theta^I} \frac{\partial}{\partial \theta^J}$$

## C. Entropic Gravity Formulation

### 1. Enhanced Entropic Force

The gravitational force emerges from entropic considerations:  $F_{\text{grav}} = T_{\text{Unruh}} \frac{\partial S}{\partial x} + F_{\text{quantum}} + F_{\text{info}}$

where:  $F_{\text{quantum}} = \frac{\hbar^2}{2m} \nabla^2 \ln \psi + \frac{l_P^2}{r^3} \left( \alpha_E + \beta_E \frac{l_P^2}{r^2} + \gamma_E \frac{l_P^4}{r^4} \right)$

$$F_{\text{info}} = k_B T_{\text{eff}} \frac{\partial I}{\partial x} \left[ 1 + \delta_I \ln \left( \frac{I}{I_0} \right) \right]$$

#### Entropic Force Parameters:

- $\alpha_E = 2.36471$  (leading quantum correction)
- $\beta_E = 0.58394$  (subleading quantum correction)
- $\gamma_E = 0.17298$  (sub-subleading correction)
- $\delta_I = 0.0847$  (information nonlinearity parameter)
- $I_0 = k_B \ln 2$  (reference information unit)

### 2. Modified Newton's Law

The resulting force law becomes:  $F = \frac{GMm}{r^2} [1 + \delta_{\text{quantum}}(r) + \delta_{\text{info}}(r, t) + \delta_{\text{running}}(\mu)]$

$$\text{where: } \delta_{\text{quantum}}(r) = \frac{l_P^2}{r^2} \left( \alpha_E + \beta_E \frac{l_P^2}{r^2} \right)$$

$$\delta_{\text{info}}(r, t) = \frac{k_B T(r)}{mc^2} \ln \left( \frac{S(r, t)}{S_0} \right)$$

$$\delta_{\text{running}}(\mu) = \frac{\lambda t^\gamma}{1 + \lambda t^\gamma} \ln \left( \frac{\mu}{\mu_0} \right)$$

## V. Cosmological Dynamics

### A. Enhanced Friedmann Equations

#### 1. Modified FLRW Dynamics

The evolution of the universe is governed by enhanced Friedmann equations:

$$H^2 = \frac{8\pi G_{\text{eff}}(t)}{3} \rho_{\text{total}} - \frac{k}{a^2} + \frac{\Lambda_{\text{eff}}(t)}{3} + H_q^2(a) + H_{\text{plasma}}^2(t) + H_{\text{info}}^2(t)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G_{\text{eff}}(t)}{3} [\rho_{\text{total}} + 3p_{\text{total}}] + \frac{\Lambda_{\text{eff}}(t)}{3} + \ddot{a}_q + \ddot{a}_{\text{plasma}} + \ddot{a}_{\text{info}}$$

#### 2. Quantum Corrections to Expansion

$$H_q^2(a) = \frac{\hbar^2}{4m_p^2 a^4} [C_1 - C_2 a^{-2} + C_3 a^{-4} + C_4 \ln a]$$

$$\ddot{a}_q = \frac{\hbar^2 a}{4m_p^2 a^4} [D_1 - D_2 a^{-2} + D_3 a^{-4} + D_4 a^{-1}]$$

**Quantum Correction Coefficients:**

- $C_1 = 0.123847$ ,  $C_2 = 0.037264$ ,  $C_3 = 0.0098735$ ,  $C_4 = 0.0034782$
- $D_1 = 0.172639$ ,  $D_2 = 0.054783$ ,  $D_3 = 0.0134729$ ,  $D_4 = 0.0067384$

#### 3. Plasma and Information Contributions

$$H_{\text{plasma}}^2(t) = \frac{8\pi G}{3} \rho_{\text{plasma}}(t) \left[ 1 + \alpha_p \left( \frac{T_e}{T_0} \right)^{\beta_p} \right]$$

$$H_{\text{info}}^2(t) = \frac{2\pi}{3} \frac{I(t)}{t_P^2} \left[ 1 + \gamma_I \ln \left( \frac{I(t)}{I_0} \right) \right]$$

where:

- $\alpha_p = 0.234$ ,  $\beta_p = 1.67$  (plasma correction parameters)
- $\gamma_I = 0.0847$  (information correction parameter)
- $T_e$  is electron temperature,  $T_0 = 2.725$  K (CMB temperature)

### B. Dark Sector Dynamics

## 1. Unified Dark Energy Model

Dark energy is described by a